Nishal Stanislaus Silva, *PhD*

Machine Learning Research Scientist | Deep Learning, Generative AI, Time-Series | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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18th October 2026

Re: Research Machine Learning Scientist – TD

Dear Hiring Manager,

I am applying for the Research Machine Learning Scientist role at Layer 6. I have a PhD (University of Trento, 2025), 10+ years across industry and academia, and a publication record in applied deep learning. I am drawn to Layer 6 because it puts research into production with short turnaround, which is how I have worked since my first role: I build ML systems end-to-end and ship them.

Three parts of my work match your requirements. For applied research with publications, I have 10+ peer-reviewed papers, including a Journal of the Audio Engineering Society article (2026) on a real-time pattern detection model that runs in 14 ms on a Raspberry Pi 4 at F1 0.76, 74x faster than the DTW baseline. For deep learning at scale, I designed benchmark suites with baselines and ablations, trained models in PyTorch and TensorFlow, and am currently fine-tuning MusicGen with LoRA across two GPUs on my own 2,276-recording dataset. For financial data, I designed and deployed ML and statistical pipelines for forensic analytics and compliance monitoring on large structured enterprise datasets at Forestpin, and I led production ML at MAS Holdings that raised operational efficiency by up to 300%.

I should be direct about the gaps. I have not worked with Spark, Azure or reinforcement learning for agent training, and my LLM work is fine-tuning and RAG rather than production post-training. My data work has been in SQL, Snowflake and Airflow on AWS, and I expect to pick up these tools quickly, as I did with each stack I have moved across.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my research and engineering experience can support Layer 6 at TD. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD